

Hands-on: Accelerating k-NN with Rcpp

Overview

In this 30 min session you will:

1. Download and inspect the C++ implementation for k-NN.
 2. Compile and load the Rcpp code.
 3. Run benchmarks comparing pure-R vs Rcpp predictions.
 4. Analyze how sample size and dimensionality affect performance.
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Setup

Download the C++ and helper scripts into your working directory:

```
1 curl -O https://raw.githubusercontent.com/mmadoiat/WSoRT/refs/heads/main/src/knn_pred.cpp
2 curl -O https://raw.githubusercontent.com/mmadoiat/WSoRT/refs/heads/main/runthis.R
```

Open the files in your editor to review the code:

- **knn_pred.cpp** contains the `knn_pred_cpp()` function (Rcpp).
 - **runthis.R** sources both R and C++ implementations and runs `microbenchmark()`.
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1. Compile the C++ code

In an R console or RStudio, run:

```
1 Rcpp::sourceCpp("knn_pred.cpp")
```

If successful, you should see `knn_pred_cpp` available:

```
1 ls("package:base") # confirm knn_pred_cpp is loaded
2 # [1] "knn_pred_cpp"
```

2. Inspect the runner script

Open `runthis.R`, which contains:

```
1 source("R/knn_s3_formula.R")           # loads knn_s3 and predict()
2 Rcpp::sourceCpp("src/knn_pred.cpp")     # loads Rcpp function
3
4 # Simulate data and benchmark
5 data <- simulate_knn_data(n = 1000, p = 5, m = 200, k = 10)
6 mb <- microbenchmark::microbenchmark(
7   Rcpp = knn_pred_cpp(data$train_x, data$train_y, data$test_x, data$k),
8   R    = knn_pred_R(data$train_x, data$train_y, data$test_x, data$k),
9   times = 20
10 )
11 print(mb)
```

Try running this script:

```
1 source("runthis.R")
```

3. Vary parameters

Modify `runthis.R` or re-run interactively to examine different settings:

- Increase **n** (training size) from 1000 to 5000 or 10000.
- Increase **p** (dimensions) from 5 to 20 or 50.
- Observe how the Rcpp version scales relative to pure-R.

Focus on how the Rcpp implementation stays much faster as complexity grows.

4. Discussion

- Where does Rcpp help most?
 - Are there settings where pure R is sufficient?
 - How might you further optimize (e.g., using STL `partial_sort`)?
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Next steps

- Try integrating this into your `knn_s3` class and Shiny app.
- Explore parallel Rcpp implementations (OpenMP).
- Consider other statistical routines with nested loops.